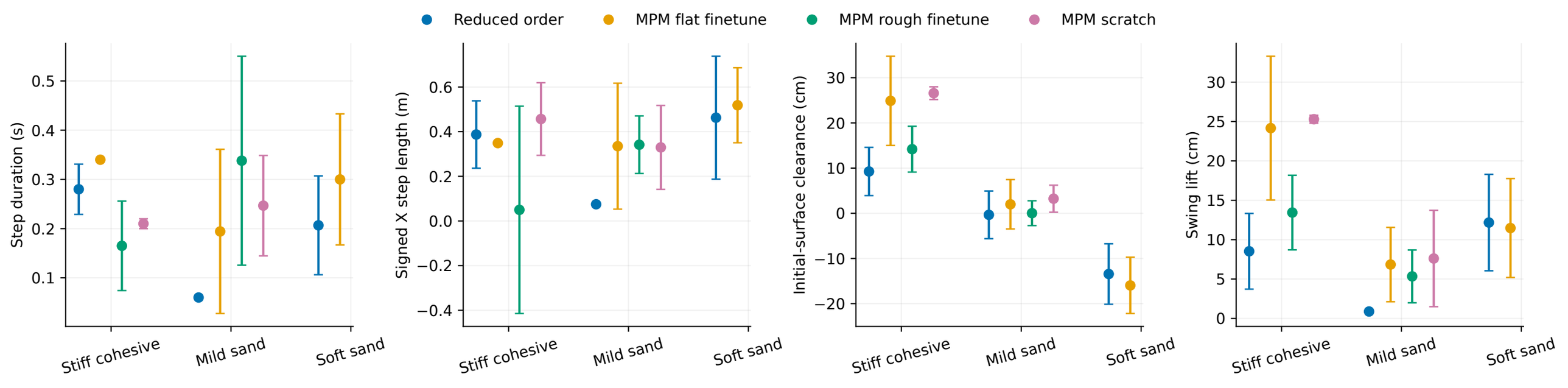
Reduced order
MPM flat finetune
MPM rough finetune
MPM scratch
Step duration (s)
0.5
0.4
0.3
0.2
0.1
Signed X step length (m)
0.6
0.4
0.2
0.0
−0.2
−0.4
Initial-surface clearance (cm)
30
20
10
0
−10
−20
Swing lift (cm)
30
25
20
15
10
5
0
Stiff cohesive
Mild sand
Soft sand